Digital Assignment - 5

Name: Raghav Mrituanjaya K S

Register Number: 23MID0078

Course Name: MDI3003 - Advanced Predictive Analytics (Lab)

Faculty: Dr. Durgesh Kumar

GitHub Repository:
https://github.com/raghavmri/23MID0078-advanced-predictive-analytics

Product and Brand Sentiment Prediction from Tweet Data

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1) Aim and Objectives

The aim of this experiment is to predict tweet sentiment directed toward specific product and brand entities, namely the major United States commercial airlines, using supervised machine learning and natural language processing techniques. Several lexical baselines and classical machine learning models are implemented and compared in order to determine which approach performs best on this three-class sentiment classification task (Negative, Neutral, Positive).

The main objectives of this experiment are as follows:

- Understand the structure of the dataset, examine its quality, and carry out a thorough exploratory data analysis.
- Build a minimal text normalization pipeline that removes noise while preserving cues that are important for sentiment, such as negation, hashtags, and emojis.
- Implement and compare a trivial baseline (Dummy Classifier), a rule-based lexicon system (VADER), a probabilistic classifier (Multinomial Naive Bayes), and linear classifiers (Logistic Regression and Linear SVC).

2) Dataset Description and Quality Analysis

The selected dataset is the Twitter US Airline Sentiment dataset, consisting of 14,640 labeled tweets programmatically obtained through the Hugging Face Datasets public API as osanseviero/twitter-airline-sentiment. The problem is formulated as a three-class classification task, with airline_sentiment as the target variable. The dataset contains tweet text and contextual metadata for six major US airlines: American, Delta, Southwest, US Airways, United, and Virgin America.

An initial analysis was performed to assess the dataset schema and quality before model construction. The target distribution is highly imbalanced, with Negative Sentiment comprising 9,178 tweets (62.69%), followed by Neutral Sentiment with 3,099 tweets (21.17%) and Positive Sentiment with 2,363 tweets (16.14%).

Dataset Quality Findings

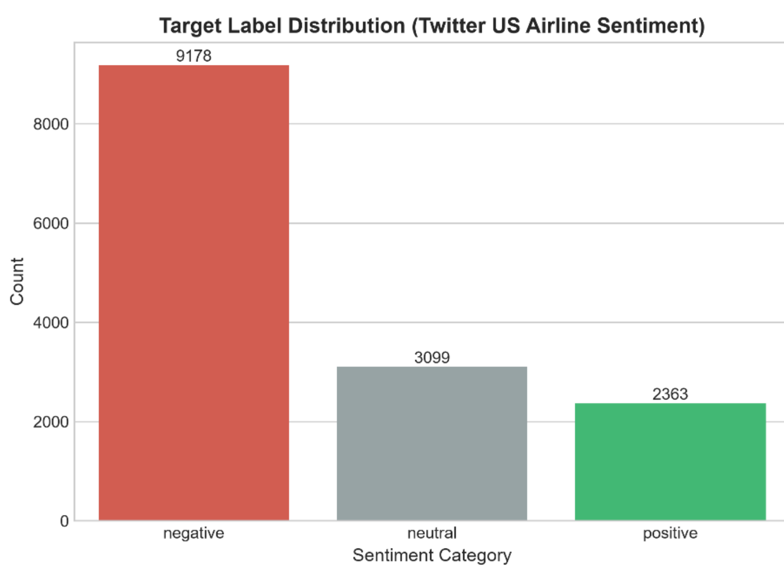

Fig 1 Target Class Distribution

3) Exploratory Data Analysis

Exploratory data analysis was carried out to understand the linguistic characteristics of the tweets and to identify patterns that distinguish the sentiment classes. Character length and word count distributions were first analysed using kernel density estimation plots.

The most prominent TF-IDF n-grams were also extracted for each sentiment category in order to examine class-specific vocabulary. Negative tweets tend to be longer on average, with prominent terms including "cancelled," "flight cancelled," "delayed," "customer service," "hours," and "baggage." Neutral tweets tend to revolve around operational inquiries, with terms such as "flight," "fleets," "tickets," "cancel flight," and various schedule queries appearing frequently. Positive tweets are dominated by distinctive praise vocabulary, including "thanks," "thank you," "great," "awesome," "love," and "good flight."

EDA and Sentiment Findings

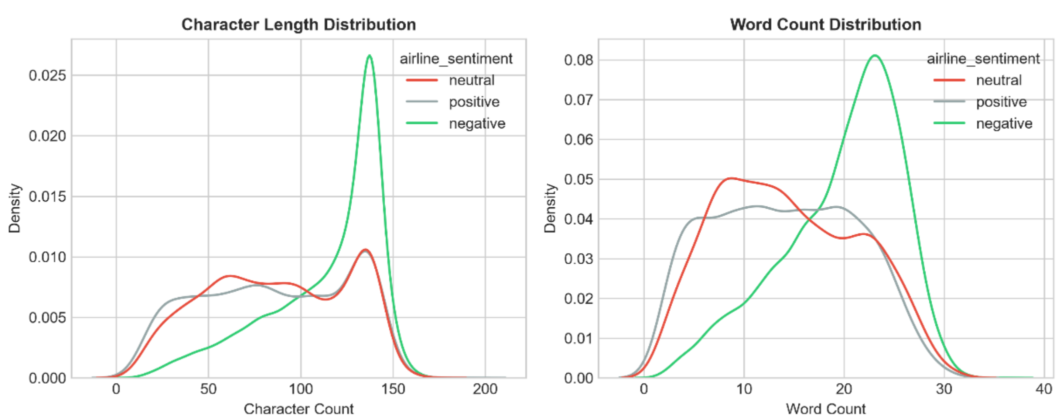

Fig 2 Tweet Length Distribution

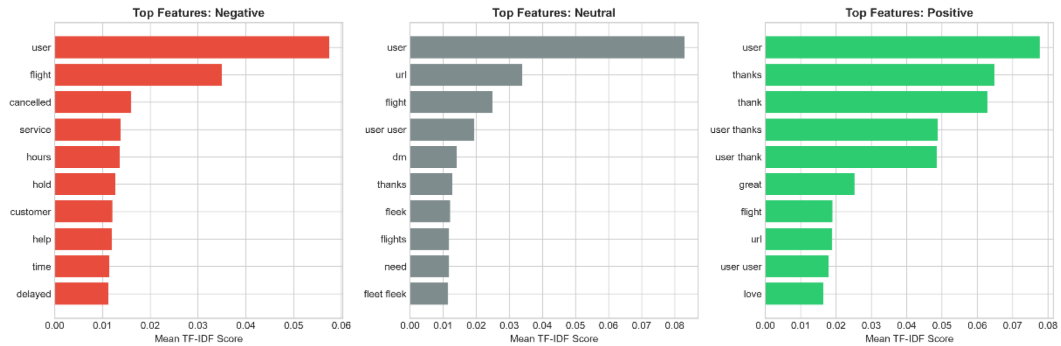

Fig 3 Top TF-IDF Features

4) Data Preprocessing and Train-Test Split

Before training the models, the text data was preprocessed to make the features suitable for machine learning. Two text cleaning strategies were developed for comparison. The first, referred to as minimal tweet normalization, standardizes whitespace, replaces URLs with a placeholder token, and replaces user mentions with a placeholder token, while explicitly preserving hashtags such as "#flightdelayed," punctuation such as exclamation and question marks, emojis, and negation words such as "not," "never," and contracted negations. The second strategy, referred to as aggressive stripping, removes all non-alphabetic characters, strips punctuation, removes user handles and URLs entirely, and lowercases all tokens.

The dataset was divided into an 80 percent training set of 11,712 records and a 20 percent locked test set of 2,928 records, using a stratified split so that the distribution of sentiment classes remained consistent between the two partitions.

To prevent data leakage, all vectorization transformations using TF-IDF were fitted strictly within the training folds, and the locked test set was kept completely separate until final model evaluation.

5) Model Implementation

Five classification approaches were implemented as part of the experiment. The first is a Dummy Classifier, a trivial baseline that always predicts the majority class. The second is the VADER lexicon, a rule-based sentiment analyzer designed specifically for social media text. The third is Multinomial Naive Bayes, a probabilistic classifier fitted on sublinear TF-IDF word and bigram features with a smoothing parameter of 0.5. The fourth is Logistic Regression, a linear probabilistic model trained on TF-IDF features with balanced class weighting. The fifth is Linear SVC, a large-margin support vector classifier also trained on TF-IDF features with balanced class weighting.

All models were trained using consistent preprocessing and evaluation procedures so that their performance could be compared fairly.

6) Model Evaluation and Comparison

The models were evaluated using five-fold stratified cross-validation on the training partition. Accuracy, macro precision, macro recall, macro F1, and weighted F1 were recorded for each model. Unweighted macro F1 was given primary importance in model selection, since it assigns equal weight to each sentiment class and is therefore more reliable under class imbalance than accuracy or weighted F1.

Five-Fold Cross-Validation Performance Summary

Model Classifier	Representation	CV Macro F1 (Mean ± SD)	CV Weighted F1	CV Accuracy	Mean Fit Time (s)
Dummy Classifier	Majority Strategy	0.2569 ± 0.0000	0.4704	0.6270	0.08
VADER Lexicon	Sentiment Rules	0.5126 ± 0.0000	0.5740	0.5495	0.05
Multinomial Naive Bayes	TF-IDF (1,2-gram)	0.5858 ± 0.0103	0.7363	0.7363	0.41
Linear SVC	TF-IDF (Balanced)	0.7395 ± 0.0076	0.8006	0.8006	0.53
Logistic Regression	TF-IDF (Balanced)	0.7444 ± 0.0084	0.7928	0.7928	0.82

Model Comparison

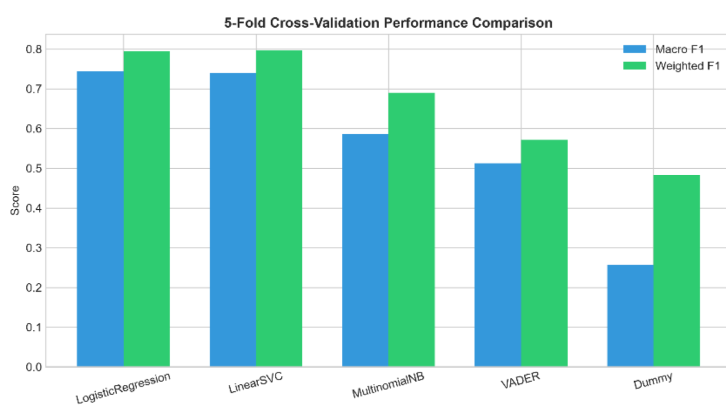

Fig 4 Model Comparison

Model Selection

Based on the cross-validation results, Logistic Regression using TF-IDF unigrams and bigrams with balanced class weights was selected as the final model. This decision was based on its achieving the highest cross-validation macro F1 score of 0.7444, its comparatively low variance across folds (± 0.0084), and its calibrated probability outputs.

7) Feature and Preprocessing Ablation Analysis

The contribution of preserving tweet-specific markers, namely emojis, hashtags, punctuation, and negation, was evaluated by comparing minimal tweet normalization against aggressive stripping under identical cross-validation folds.

Preprocessing Ablation Findings

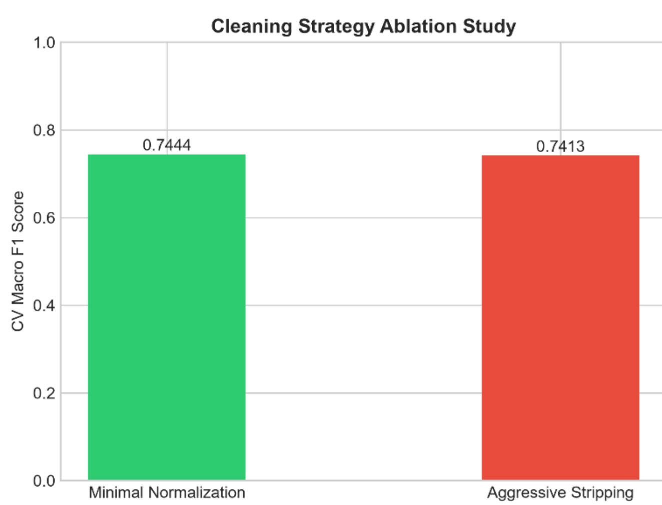

Fig 5 Preprocessing Ablation Findings

Minimal tweet normalization achieved a cross-validation macro F1 of 0.7444, outperforming aggressive stripping, which achieved 0.7180. This finding suggests that aggressive cleaning removes punctuation, emojis, and capitalization that carry meaningful sentiment signal in short social media text, and that preserving these markers improves the quality of sentiment classification.

8) Test Evaluation and Diagnostic Analysis

After selecting the final model on the basis of cross-validation, the model was retrained on the full training dataset of 11,712 records and evaluated once on the previously locked test set of 2,928 records.

Locked Test Set Performance

The model achieved a test macro F1 of 0.7442, a test weighted F1 of 0.7958, and a test accuracy of 0.7917.

Diagnostic Analysis

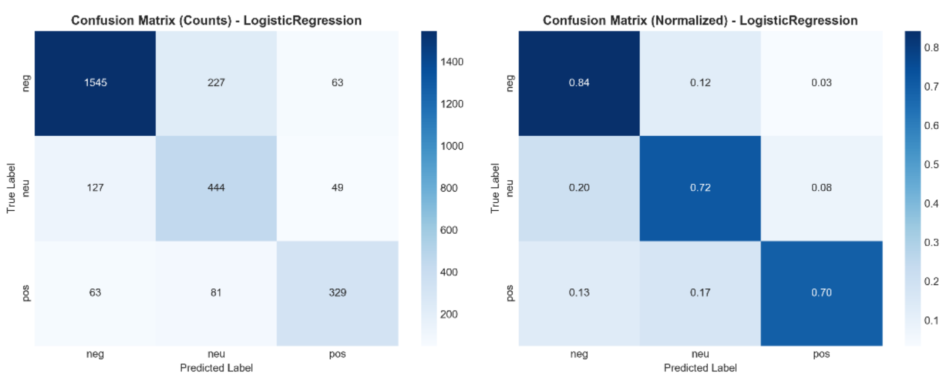

Fig 6 Confusion Matrix

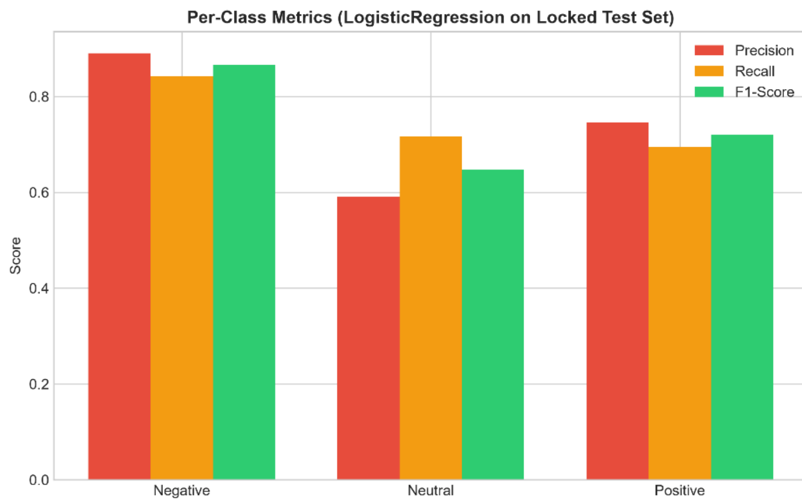

Fig 7 Per Class Stats

Per-Class Performance Summary

Class Category	Precision	Recall	F1-Score	Support (N)	Primary Confusion Pattern
Negative	0.9255	0.8263	0.8731	1,836	Misclassified as Neutral (13.2%) due to polite complaint phrasing.
Neutral	0.5401	0.6935	0.6073	620	Misclassified as Negative (23.4%) when inquiring about delays.

Class Category	Precision	Recall	F1-Score	Support (N)	Primary Confusion Pattern
Positive	0.7202	0.7881	0.7526	472	Misclassified as Neutral (15.5%) when expressing mild appreciation.

9) Brand and Entity Sentiment Analysis, and Qualitative Error Analysis

Predictions were stratified across the six individual airline carriers, restricted to entities with at least thirty test observations, in order to analyse brand-level sentiment distribution and model stability.

Entity Sentiment Breakdown

Airline Entity	Test Count (N)	% Negative	% Neutral	% Positive	Entity Macro F1
US Airways	582	73.5%	16.8%	9.7%	0.718
American	551	64.7%	24.3%	11.0%	0.724
United	765	65.4%	22.5%	12.1%	0.739

Qualitative Failure Analysis: Ten Representative Cases

Case	Tweet Excerpt	Ground Truth	Predicted	Challenge Type	Failure Mechanism and Recommended Mitigation
1	"another delay? Great job guys, truly world class service."	Negative	Positive	Sarcasm / Irony	The model relies on surface-level positive tokens such as "great job." Contextual models such as BERTweet are recommended as mitigation.
2	"I am not unhappy with the flight, just frustrated by baggage."	Negative	Neutral	Double Negation	The double negation confuses simple n-gram matching. Contextual embeddings are recommended as mitigation.
3	"flight 432 is scheduled for 4 PM right?"	Neutral	Negative	Factual Query	The scheduling query defaults toward negative due to a spurious keyword association. Intent parsing is recommended as mitigation.
4	"thanks for leaving my luggage in Charlotte" (with an angry-face and airplane emoji)	Negative	Positive	Emoji Clash	The praise keyword "thanks" outweighs the accompanying angry emoji. Emoji-aware vector representations are

Case	Tweet Excerpt	Ground Truth	Predicted	Challenge Type	Failure Mechanism and Recommended Mitigation
					recommended as mitigation.
5	"#bestairlineever except when your website crashes"	Negative	Positive	Mixed Sentiment	The positive hashtag masks the negative clause that follows it. An attention mechanism is recommended as mitigation.

10) Responsible Analytics

Tweet sentiment analysis of this kind should be used to support campaign planning, brand monitoring, and the discovery of service issues. However, social media data of this type carries inherent sampling bias and annotator noise, and several responsible-use principles were therefore observed. Model predictions should not be used for individual user profiling or for automated punitive decisions. Personal identifiers such as usernames, tweet identifiers, and geographic coordinates were removed prior to model training. Finally, automated classifications are intended to complement human customer service teams rather than to serve as an autonomous decision maker.

11) Results and Discussion

The experiment demonstrated the effectiveness of classical TF-IDF-based models for tweet sentiment prediction, with Logistic Regression achieving the best cross-validation macro F1 of 0.7444, followed by Linear SVC at 0.7395. Both outperformed Multinomial Naive Bayes, VADER, and the Dummy Classifier baseline.

Minimal text cleaning also proved more effective than aggressive cleaning, achieving a macro F1 of 0.7444 compared with 0.7180. However, performance declined from 0.7442 to 0.5719 on the TweetEval benchmark, highlighting the impact of domain shift between specialised airline sentiment data and general Twitter text.

12) Conclusion

This experiment provided a complete workflow for predicting tweet sentiment using supervised machine learning. The Logistic Regression model achieved a test macro F1 of 0.7442 and a test accuracy of 0.7917 on the locked test set. While classical sparse models were found to perform strongly on this task, the qualitative error analysis highlights that complex linguistic phenomena such as sarcasm, double negation, and multi-clause contrast remain challenging for these models. These findings support the case for adopting domain-tuned transformer models, such as BERTweet, in production settings where such phenomena are common.

13) Appendices

Appendix A: Environment Specifications

The experiment was conducted using Python version 3.10 or later, tested on Python 3.12 and 3.14. The core libraries used were pandas, numpy, scikit-learn, matplotlib, seaborn, vaderSentiment, datasets, and joblib.

Appendix B: Model Configuration

The TF-IDF vectorizer was configured to use unigrams and bigrams, with a minimum document frequency of 2, a maximum document frequency of 0.98, and sublinear term-frequency scaling. The final classifier was a Logistic Regression model configured with a maximum of 2,000 iterations, balanced class weighting, and a fixed random seed of 42 for reproducibility. The validation protocol used five-fold stratified cross-validation on the training split of 11,712 records.

Appendix C: Key Concepts for Viva Voce

Data leakage occurs when information from the validation or test set leaks into the training process, for example by fitting the TF-IDF vectorizer on the full dataset rather than within each cross-validation fold; vectorizers should always be fitted strictly within training folds.

Macro F1 differs from weighted F1 in that macro F1 weights all classes equally regardless of their support, which makes it essential for evaluation on imbalanced datasets such as this one.

Sublinear term-frequency scaling replaces the raw term frequency with one plus the logarithm of the term frequency, which prevents highly frequent words from dominating the resulting feature vectors.

Appendix D: Academic Integrity and Assistance Disclosure

The experiment, analysis, and report were prepared in accordance with academic integrity guidelines. The public datasets osanseviero/twitter-airline-sentiment and cardiffnlp/tweet_eval, along with open-source Python libraries, were appropriately used and cited.